QUESTION 1

Database is where we store data, databases are essential because information is easier to structure and maintain. Data can be searched and retrieved much faster than in ad hoc files. for example online stores: customer, carts, order etc
Banking system: accounts, balances and transactions

QUESTION 2

A data warehouse aggregates data from various sources into a central store optimized for querying and analysis. It typically uses extract, transform and load (ETL), processes to clean, prepare and organize data for business intelligence (BI) and other data analytics use cases.

QUESTION 3

A data lake is a repository designed to store large volumes of raw data, typically using low-cost cloud object storage. This approach allows the ingestion and storage of structured, semi-structured and unstructured data with a single platform.
Unlike databases and data warehouses, data lakes don't enforce strict schema's and today's data lakes uses affordable, scalable cloudstorage making them ideal for large amounts of diverse data.
The key difference is that data warehouses typically clean and prepare data before ingestion so that it is immediately ready for analytics. Warehouses are optimized for structured data.

QUESTION 4

A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area. Data marts can improve team efficiency, reduce cost and facilitate smarter tactical decision-making in enterprises.
A data mart warehouse is built from an existing data warehouse or other data sources through a complex procedure that involves multiple technologies and tools to design and contract a phisical database, populate it with data, and set up intricate access and management protocols.

QUESTION 5

Structured data organises information in standardized format defined by a schema. It usually stores the data in tables with rows and columns. eg Data in SQL database or other relational systems. It is easy to search and analyze with standard tools.

Unstructured data does not have a standardized format or data model. Systems store unstructured data in its native formal and there are many different types. eg text files, photographs, video and audio recording.

Semi-structured data falls between structured and unstructured data. Some of it follows a standardized format while the rest does not. tags are another example of data your organization often treats as semi-structured e.g teams may add tags to real-time data to make it easier to use and analyze.

QUESTION 6

Data modelling is the process of analyzing and defining all the different data types your business collects and produces, as well as the relationships between those bits of data. By using text, symbols, and diagrams, data modelling concepts creates visual representations of data as it's captured, stored and used at your business. As your business determines how data is used and when the data modelling process becomes an exercise in understanding and clarifying your data requirements.

QUESTION 7

Data mining is the use of machine learning and statistical analysis to uncover patterns and other valuable information from large data sets. Business analyze customer behaviour and browsing patterns to increase retention, identify at-risk customers and maximize marketing ROI. For example

Retail: market basket analysis helps retailers analyze customer purchase patterns to enhance product placement and cross-selling opportunities.

Healthcare: Disease diagnosis and predictions using patient Data employs algorithms to predict diseases from patient records, improving early diagnosis and treatment plans.

QUESTION 8

Database management system (DBMS) is a software technology that manages the information in your database. A DBMS makes storing data easy while increasing availability, reliability and performance. it also provides tools to identify data correlations and perform analytics within the system as needed.

PostgreSQL: is primarily a relational database management system with object-oriented features, including support for object programming languages, object, classes and inheritance

MySQL: is an open-source Relational Database Management System (RDBMS) tha uses SQL to manage and manipulate data. It is widely used for storing, retrieve and managing structured data in applications ranging from small personal projects to large-scale enterprise systems.

QUESTION 9

ETL: is extract, transform, Load is a data integration process that combines, cleans and organizes data from multiple sources into a single, consistent data set for storage in a data warehouse, data lake or other target system.

Transform 2

~~Extract~~: In a staging area, the raw data undergoes data processing. Here, the data is transformed and consolidated for its intended analytical use case. The process includes filtering cleaning, aggregating, de-dublicating, validating and authenticating the data.

Extract: raw data is exported from source locations to a staging area. Data management system teams can extract data from a variety of different sources, which can be structured or unstructured. SQL, CRM

LOAD: In the last step, the transformed data is moved from the staging area into a target data warehouse.

QUESTION 10

CSV: structured file formats used for tables, rows and columns (great for spreadsheets and databases). it is used when working with clean, table-style business data.

JSON: Semi-structured format, data has structure but it is flexible (commonly for APIs and web apps) It is used when ingesting data from application, APIs or cloud system.

Parquet: Optimized for large-scale analytics (spark, snowflake, Databricks) column-based storage highly efficient row-based good for streaming and compression used when working with large volumes of data and distributed system.

QUESTION 11

Artificial intelligence is a technology that enables computer and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy

Narrow AI (weak AI) systems that are designed and trained for a specific task or narrow range of tasks. An example of weak AI, virtual personal assistant like siri, Alexa while

General AI (strong AI) refers to a level of artificial intelligence that matches human intelligence in understanding, learning and solving. Unlike narrow AI, which excels at specific tasks like facial recognition or language translation, General AI possesses the ability to perform a wide range of cognitive tasks with adaptability and versatility. Its development aims to replicate the dynamic and flexible problem-solving capabilities of the human mind, opening new horizons for innovation.

QUESTION 12

Machine learning is the subset of artificial intelligence (AI) focused on algorithms that can 'learn' the patterns of training data and subsequently, make accurate inferences about new data. This pattern recognition ability enables machine learning models to make decisions or prediction without explicit, hard-coded instructions.

Types of machine learning

Supervised learning: trains a model to predict the correct output for a given input. it applies to tasks that require some degree of accuracy relative to some external ground truth, such as classification or regression.

Unsupervised learning: trains a model to discern intrinsic patterns, dependencies and correlations in data. Unlike in supervised learing, unsupervised learning tasks don't involve any external ground truth against which its output should be compared.

Reinforcement Learning: trains a model to evaluate its environment and take an action that will garner the greatest reward. RL scenarios dont entail the existence of a singular ground truth, but they do entail the existence of good and bad actions.

QUESTION 13

Deep Learing is a subset of machine learning driven by multilayered neural network whose design is inspired by the structure of the human brain. Deep learing models power most state-of-the-art artificial intelligence AI today, from computer vision and generative AI to self-driving cars and robotics.

Neutral networks and deep learning have become inextricably associated with one another, they are not strictly synonymous; deep learning refers to the training of models with atleast 4 layers (though modern network architures are often much deeper

than that)

QUESTION 14

Natural language processing (NLP) is a subfield of computer science and artificial intelligence (AI) that uses machine learning to understand and communicate with human language.

NLP enables computers and digital devices to recognize, understand and generate text and speech by combining computational linguistics, the rule-based modelling of human language together with statiscal modelling, machine learning and deep learning.

QUESTION 15

Large language models (LLMs) are a category of deep learing models trained on immense amounts of data, making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks. LLMs are built on a type of neural network architecture called a transformer which excels at handling sequences of words and capturing patterns in text.

LLMs work as a giant statistical prediction machines that repeatedly predict the next work in a sequence. They learn patterns in their in their text and general language that follows those patterns.

QUESTION 16

Generative AI, sometimes called gen AI, is artificial intelligence (AI) that can create original content such as text, images, video, audio or software code in response to a user's prompt or request. Generative AI relies on sophisticated machine learning models called deep learning algorithms that simulate the learning and decision-making processes of the human brain.

~~CHAT~~

ChatGPT; is an open source AI powered natural language processing tool that enables users to have human like conversations with a chatbot.

Google Bard; is a powerful large language model (LLM) chatbot by Google AI that can answer questions or offer prompts on range of different topics. It can generate different forms of text-based content, such as poems, code, scripts, musical pieces, emails or letter as well as summarize text and translate between languages.

QUESTION 17

A Training Dataset refers to the dataset used to create a representative model by inferring from the data. It is a database with known attributes and target that is used to train the model. All machine learning starts with a data set, or a collection of data. A dataset could be made up of spreadsheets, video footage, web pages, PDFs or any other type of data.

Generally speaking the more training data that is fed into a model, the better the model's performance.

Incorrect training data could lead to biased outputs that might unfairly represent or otherwise discriminate against certain groups or individuals. These impacts erode AI and organizations that use AI. They can also lead to legal and regulatory penalties for businesses.

QUESTION 18

Computer vision is a subfield of artificial intelligence (AI) that equips machines with the ability to process, analyze and interpret visual inputs such as images and video. It uses machine learning to help computers and other systems derive meaningful information from visual data

Data gathering: collect the necessary visual data. Hospitals generate huge volumes of chest X-rays, which they can use to train a computer vision algorithm. Since the goal is for the algorithm to classify whether an X-ray image depicts pneumonia or not, hospitals will need to compile a dataset of chest X-rays scans and correctly label or annotate each scan as either normal or signifying pneumonia.

Preprocessing: An AI model is only as good as the data used to train, which makes high-quality data crucial for computer vision. Preprocessing can help improve data quality through data cleaning and enhancements like adjusting brightness or contrast to sharpen images, as well as resizing and smoothing.

QUESTION 19

An API, or application programing interface, is a set of rules or protocols that enables software applications to communicate with each other to exchange data, features and functionality. Its useful to think about API communication in terms of a request and response between a client and a server. The application submitting the request is the client, and the server provides the response. The API is the bridge establishing the connection between them.

Q 19

For example, third-party payment processing. purchases a product on an e-commerce site, the site might prompt the user to Pay with PayPal or another type of third-party system. This function relies on APIs to make the connection.

QUESTION 20

An API key is a unique identifier used to authenticate software and systems attempting to access other software or systems via an application programming interface, or API.

An application programming interface is a set of rules or protocol that enable software to communicate with each other to exchange data, features and functionality.

API keys enable an API server to identify which software and applications are sending a call to the API. The generated key is used for every request made from that application or project to the associated API (until the key is refreshed or deleted). If they are exposed publicly they could become vectors for a cyber attack.

QUESTION 21

A REST API is an application programming interface (API) that conforms to the design principles of the representation state transfer (REST) architectural style, a style used to connect distributed hypermedia systems. REST APIs are sometimes referred to as RESTful APIs or RESTful web APIs.

HTTP methods (GET, POST, PUT, DELETE)

GET: The HTTP GET method retrieve a resource

POST: The POST method creates new resources.

PUT: is used to update or create a resource. It sends the comple resou in the request body and replaces the existing one at the specified URL.

DELETE: It is used to delete a resource identified by a URI. On successful deletion, return HTTP status 200 (OK) along with a response body.

QUESTION 22

JSON (JavaScript object Notition) is a lightweight format for storing and transporting data, often used when data is sent from a server to a web page.

Its simplicity, flexibility and wide compatibility make it the go to choice for developers building APIs. Its simplicity is crucial for improving productivity and reducing errors.

QUESTION 23

Prompt engineering helps generative AI models better comprehend and respond to a wide range of queries, from the simple to the highly technical.

Prompt engineers play a pivot role in crafting queries that help generative AI models understand not just the language but also the nuance and intent behind the query. A high quality, through and knowledgeable prompt, in turn, influences the quality of AI-generated content, whether its images, code, data summaries or text. Poorly written prompt example 'Analyze this' the AI does not know the kind of analysis you want.

QUESTION 24

MAP function returns an array formed by mapping each value in the array(s) to a new value by applying a LAMBDA to create a new value.

Python map() function

```
s = ['1', '2', '3', '4']
res = map(int, s)
print(list(res))
```

QUESTION 25

Cloud computing is on-demand access to computing resources physical or virtual servers, data storage, networking capabilities, application development tools, software, AI-powered analytic platforms and more over the internet with pay per use pricing.

Three main cloud service models

PaaS: Platform-as-service-provides software with as an on-demand platform-hardware, complete software stack, infrastructure and development tools for running, developing and managing applications without the cost, complexity and inflexibility of maintaining that platform on-premises

IaaS: Infrastructure-as-a Service provide access to fundamental computing resources-physical and virtual servers, networking & storage-over the internet on a pay-as-you-go basis.

SaaS: Software as a service also known as cloud-based software or cloud applications, is interative application software hosted in the cloud.

QUESTION 26

Version Control system(VCS) give software engineering teams complete visibility to the code history and a single source of documention for all files, folders and messages.

Q 26

Git is a tool used to keep track of changes to files, especial the code of the projects. It is termed a distributed version control system because of its behaviour to allow multiple people to work on the same project, even if they are not connected to a commo server.

Commit: Everytime you make changes and save them in Git, these are called commits. The repository keeps track of all the commits you have made.

Branches: A repository allows you to create Branches and work on new features and fixes.

QUESTION 27

Jupyter Notebooks are an open document format based on JSON. They contain a complete record of the user's session and include code, narrative text, equation and rich output.

QUESTION 28

Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high level built in data structure, combined with dynamic typing and dynamic binding make it attractive for Rapid Application Development, as well as as a scripting language to connect existing components together.

Its simplicity, versatility and robust support system have made it almost indispensable for data scientists, with Python now appearing as a requirement in nearly every job advertisem in ~~nearly~~ ~~every~~ the field.

Python Libraries
- Pandas
- Numpy
- matplotlib
- Scikit-learn

QUESTION 29

Data privacy, also called information privacy, is the principle that a person should have control over their personal data, including the ability to decide how organization collect store and use their data. Supporting data privacy means taking steps like obtaining user consent before processing data, protecting data from misuse and enabling users to actively manage their data.

Q 29

Many organization have a legal obligation to uphold data privacy under laws like the General Data Protection Regulation (GDPR), is a set of EU rules on data protection and privacy.

QUESTION 30

Data ethics refers to the principles behind how organization gather, protect and use data. Its a field of ethics that focuses on the moral obligations that entities have (or should have) when collection and disseminating information about us. In a world where data is more valuable and ubiquitous than ever, data ethics issues are more pressing now that at any time in history.

unethical use of data or AI

- Attack vector: They may use AI technologies as tools facilitate crimes such as a automated cyberattacks, fraud or identity theft
- Attack surface: They may abuse existing AI application by attacking them or breaking them intentionally such as hacking into smart home assistants like Amazon's Alexa.

QUESTION 31

Data governance is the data management discipline that focussed on the quality, security and availabity of an organizat data. Data governance helps ensure data integrity and data security by defining and implementing policies, standards and procedures for data collection, ownership, storage, processin and use.

Key components of a good data governance framework

- Program goals, roles and duties: it defines specific goal or set of goals, such as enhancing data quality, supporting compliance or enabling data-driven decision-making.
- Data standards, policies and processes: governance frameworks set parameters around the data to be governed and the wanted outcomes. This process includes setting guidelines for data formats, data models, master data management (MDM) metadata, naming conventions and more.

QUESTION 32

AI bias, also called machine learning bias or algorithm bias, refers to the occurrence of biased results due to human biases that skew the original training data or AI algorithm leading to distorted outputs and potentially harmful outcome

When AI bias goes unaddressed, it can impact an organization success and hinder people's ability to participate in the economy and society. Bias reduces AI's accuracy, and therefor its potential.

QUESTION 33

	Data Analyst	Data Scientist	Data Engineer
Roles	• collect, cleans, stores and organises data.	• develops and implement data-driven solutions to overcome business challenges.	• builds and maintains the data infrastructure other data team members use to perform various task.
Skills	• SQL knowledge, excel skills, critical thinking	• Coding skills, data handling, problesolving, strong math basics	• Programming skills understanding databases, cloud platform knowledge
Tools	• Mircosoft excel/ Google sheet, SQL, PowerBI	• Jupyner Notebook, R, Tableau, Python	• Apache Hadoop, Apache Spark, SQL Databes, Air-flow, AWS

QUESTION 34

Business intelligence (BI) is a set of technological processes for collecting and analyzing organization data to yield insights that inform business strategies and operations.

BI tools

Power BI: is a data visualization and reporting platform that is used by business and professionals every day. it commonly used by business analysts, it is also designed to be easily accessible for those without any specialized data knowledge. it is capable of turning data into insights, reports, and interactive visualizations.

Tableau: it is critical to helping businesses make better decisions by unlocking and surfacing insights with the vast amount of data they have. This data is often siloed, disjoined, and difficult to manage. With Tableu, every company can get actionable insights to drive their business forward, helping save money and be more efficient

QUESTION 35

Artificial Intelligence and Machine Learning is the most interesting because, when I first started learning data analytics, I thought the goal was to memorize everything. .e SQL syntax or every PowerBI trick. It felt like if I didnt know it off by my head I wasnt enough.

But the truth is: no serious analyst carries hundreds of formulas by head, what they carry is understanding. You have to understand what the data is saying and what problem need to be solved. And delivering insights that matter.

QUESTION 36

APIs, Tools and Developer concepts was quite difficult to understand. I have used Youtube and Tiktok for better understanding.

QUESTION 37

AI wont replace analysts, but analysts who use AI will replace those who don't. Some people are still trying to prove they can do everything alone, others are learning how to work smarter. In the longrun the second group moves faster. We have to understand something simple; tools change but thinking remains the advantage.

REFERENCES

1. IBm.com/think/topics/data-warehouse
2. Salesforce.com/new/stories/tableau
3. microsoft.com/em-us/powerplatform
4. www.techtarget.com
5. sciencedirect.com